Numerical problem:-

x_1 x_2

x = bank river

$$x_1 = \begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix} \quad \begin{bmatrix} 0.0 \\ 1.0 \end{bmatrix}$$

2x1

$$W^Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \qquad W_K = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \qquad W^V = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

Attention (Q, K, V) = → Z_1 for bank
→ Z_2 for river

2x2

Z_1 and Z_2 needs to be find out!

$$Z = \text{softmax}\left(\frac{Q K^T}{\sqrt{d_k}}\right) \times V$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix}$$

$$\begin{bmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{bmatrix}$$

multiply.

Q = X × W_Q

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0 & 0+0 \\ 0+0 & 0+2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$W_Q = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

multiply.

K = X × W_K

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+0 & 1+0 \\ 0+0 & 0+1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$K^T = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$V = X * W_V$ = Multiply.

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2+0 & 0+0 \\ 0+1 & 0+3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$Z = \text{softmax}\left(\frac{\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}{\sqrt{2}} \right) \times \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$$

$$= \text{softmax}\left(\begin{matrix} 1+0 & 0+0 \\ 0+2 & 0+2 \end{matrix} \Big/ \sqrt{2} \right) \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$= \text{softmax}\left(\begin{matrix} 1 & 0 \\ 2 & 2 \end{matrix} \Big/ \sqrt{2} \right) \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$= \text{softmax}\begin{pmatrix} 1/\sqrt{2} & 0 \\ 2/\sqrt{2} & 2/\sqrt{2} \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$\text{softmax} = \frac{e^{x_i}}{\sum_{j=1} e^{x_j}} \quad \therefore = \text{softmax}\begin{pmatrix} 0.707 & 0 \\ 1.414 & 1.414 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$\begin{bmatrix} 0.669 & 0.33 \\ 0.5 & 0.5 \end{bmatrix} = \begin{pmatrix} 2.027 & 1 \\ 4.112 & 4.112 \end{pmatrix} = \begin{pmatrix} 0.669 & 0.33 \\ 0.5 & 0.5 \end{pmatrix}$$

$$= \begin{pmatrix} 0.669 & 0.33 \\ 0.5 & 0.5 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1.338+0.33 & 0+0.99 \\ 1+0.5 & 0+1.5 \end{pmatrix} = \begin{pmatrix} 1.668 & 0.99 \\ 1.5 & 1.5 \end{pmatrix}$$